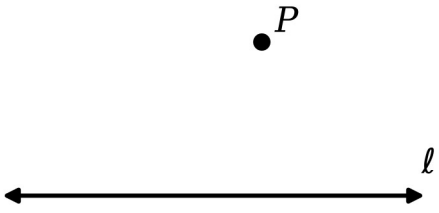
Honors Geometry – Construction Lab 4 Name: _____ Period: _____

Sections 3.1 – 3.4 — Parallel and Perpendicular Lines Materials: compass · straightedge · protractor · cm ruler

Part 1 — Construct a line through P parallel to ℓ. Compass and straightedge only. Draw a transversal through P, then copy an angle AT P. Leave your arcs showing.



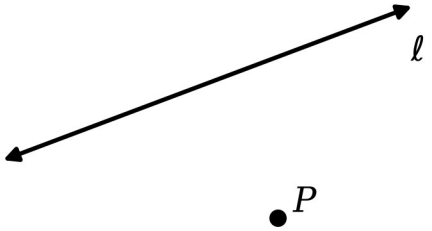
a. Copy the CORRESPONDING angle.

Your lines are parallel by the

_____

★ **Think deeper:** Your transversal could cross ℓ at ANY angle. Would you get a different parallel line? Which postulate from 3.1 answers this?

_____



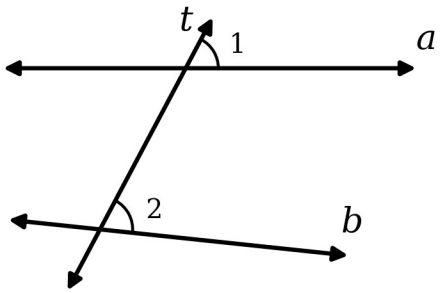
b. This time copy the ALTERNATE INTERIOR angle.

Reason the lines are parallel:

_____

★ **Think deeper:** Measure the two consecutive interior angles in your figure. What should they add to, and which theorem guarantees it?

_____



c. Dev changed his compass width partway through and got line b.

m∠1 = _____ m∠2 = _____

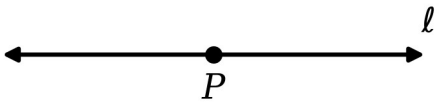
Is a ∥ b? Explain with a postulate or converse.

_____

★ **Think deeper:** Add the two interior angles on the LEFT side of t. Is the sum under or over 180°? Predict the side where a and b meet.

_____

Part 2 — Construct each perpendicular. Compass and straightedge only. Then check with a protractor and ruler.



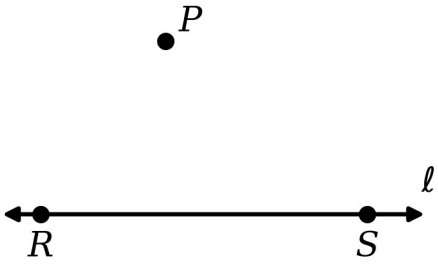
a. Construct the line through P perpendicular to ℓ.

Check: angle = _____

★ **Think deeper:** You did this exact construction in Lab 3. Which part? Why does it give EXACTLY 90°?

_____

_____



b. Construct the perpendicular from P to ℓ. Label where it meets ℓ as Q.

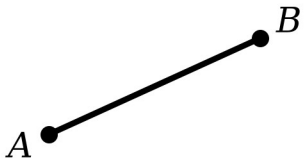
PQ = _____ cm

PR = _____ cm PS = _____ cm

★ **Think deeper:** Which segment is shortest? What is the distance from P to ℓ, and why THAT segment?

_____

_____



c. Construct the perpendicular bisector of AB. Mark any point X on it.

XA = _____ cm XB = _____ cm

★ **Think deeper:** Mark a point Y NOT on the bisector. Is YA = YB? What word describes EVERY point on the bisector?

_____

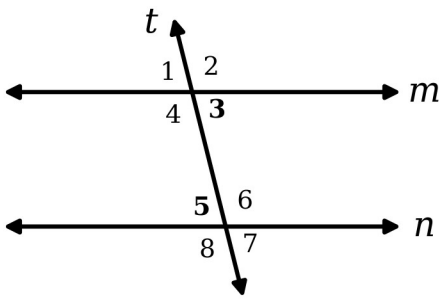
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Honors Geometry – Construction Lab 4 Name: _____ Period: _____

Sections 3.1 – 3.4 — Parallel and Perpendicular Lines

Part 3 — Parallel lines and angle algebra. Write the equation, solve for x, then find EVERY angle asked for.

a. m ∥ n. m∠3 = (4x + 12)° and m∠5 = (6x − 20)°. Find x and m∠3.



∥ + alternate interior ⇒ angles EQUAL

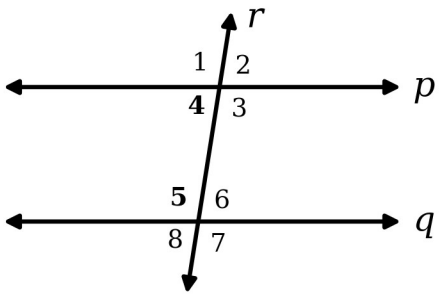
m∠3 = m∠5

★ **Think deeper:** Now find all 8 angles with only ONE more calculation. Why is one enough?

_____

_____

b. p ∥ q. m∠4 = (3x + 15)° and m∠5 = (5x − 11)°. Find x and both angles.



∥ + consecutive interior ⇒ angles ADD to 180°

m∠4 + m∠5 = 180

★ **Think deeper:** Could r be perpendicular to p? Set m∠4 = 90, solve, and test m∠5.

_____

_____

c. Ky says: "Two lines perpendicular to the same line must be perpendicular to each other." Is he right? Sketch it, then name the theorem that settles it.

_____

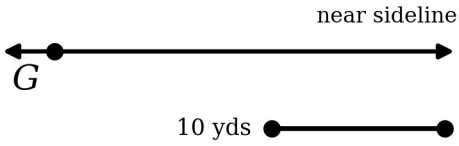
_____

★ **Think deeper:** Look at a corner of this room. Find two edges that are BOTH perpendicular to a vertical edge. What does the theorem need that the room breaks?

_____

_____

Part 4 — Team Challenge: Lay out the field. Work with your group. Compass and straightedge only — the protractor is for CHECKING at the end.



G is where the goal line crosses the near sideline. The segment shows 10 yards to scale.

Build the goal line, the 10-yard line, and (in d) the far sideline.

a. Plan first: which constructions build the goal line and the 10-yard line?

_____

_____

b. Build them. Where the 10-yard line meets the sideline: m∠ = _____

c. Why MUST the goal line be ∥ to the 10-yard line, even before you measure?

_____

★ **Think deeper:** A groundskeeper has only a tape measure. How can he check that the 10-yard line is parallel to the goal line?

_____

d. Extension: mark F on your goal line, 3 cm from G. Construct the far sideline through F, parallel to the near sideline. Write your plan before you build.

_____

_____

Why must the far sideline be ⊥ to the 10-yard line? Name the theorem.

_____

Check: m∠ = _____

★ **Think deeper:** Your field is a rectangle. Using only 3.3–3.4 theorems, explain why ALL FOUR corners are right angles.

_____